



Problem-Solution Fit

Date	2 July 2026
Team ID	XXXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Farmers, Agricultural Students, Agricultural Officers, Researchers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited internet access, limited technical knowledge, budget constraints, lack of modern farming tools.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional methods: Low cost but less accurate. Agricultural experts: Reliable but not always available. Online sources: Easy access but may provide generic recommendations.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Crop failure, low yield, changing weather	9. PROBLEM ROOT / CAUSE [RC] Lack of accurate crop prediction, changing	7. BEHAVIOR + ITS INTENSITY [BE] Farmers usually depend on personal experience,

conditions, need for higher productivity and profit	climate, insufficient knowledge about soil nutrients, and limited access to expert guidance.	neighboring farmers, or local experts. Intensity: High during crop planning seasons.
3. TRIGGERS TO ACT [TR] Before: Confused, worried, uncertain. After: Confident, satisfied, informed, hopeful.	10. YOUR SOLUTION [SL] OptiCrop – Smart Agricultural Production Engine uses Machine Learning to analyze soil nutrients (N, P, K), temperature, humidity, pH, and rainfall, then recommends the most suitable crop for cultivation, helping farmers improve productivity and reduce crop failure.	8. CHANNELS OF BEHAVIOR [CH] ONLINE OFFLINE Online: Mobile applications, agricultural websites, government agriculture portals. Offline: Agricultural officers, local markets, farmer meetings, training programs.
4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, worried, uncertain. After: Confident, satisfied, informed, hopeful.		